

Standard-normal - CDF (cumulative distribution func)

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$$

zu $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-u^2} du$

Substitution $u = \frac{t}{\sqrt{2}} \Leftrightarrow t = \sqrt{2} \cdot u$
 $dt = d(\sqrt{2} \cdot u) \Leftrightarrow \sqrt{2} du$
 $e^{-t^2/2} = e^{-(\sqrt{2} \cdot u)^2/2} = e^{-u^2}$

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{x/\sqrt{2}} e^{-u^2} \sqrt{2} du$$

$\frac{\sqrt{2}}{\sqrt{2\pi}} = \frac{1}{\sqrt{\pi}}$

$$= \frac{1}{\sqrt{\pi}} \cdot \int_{-\infty}^{x/\sqrt{2}} e^{-u^2} du \left[\int_{-\infty}^a = \int_{-\infty}^0 + \int_0^a \right] \int_{-\infty}^0 e^{-u^2} du = \frac{\sqrt{\pi}}{2}$$

Symmetrie von e^{-u^2}

$$= \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{x/\sqrt{2}} e^{-u^2} du = \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right) = \frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{2}}} e^{-u^2} du \Rightarrow \Phi = \frac{1}{2} (1 + \operatorname{erf}(\frac{x}{\sqrt{2}}))$$

$$\text{GULL}(x) = x \cdot \Phi(x)$$

$$= x' \cdot \Phi + x \cdot \Phi'$$

$$= \Phi + x \cdot \Phi'$$

$$\Phi'(x) =$$

$$\frac{1}{2} \left(1 + \text{erf} \left(\frac{x}{\sqrt{2}} \right) \right) + x \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-x^2/2}$$

* $\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$ (bekannt)
 wir brauchen nur
 $\int_{-\infty}^0 \Rightarrow$ also Hälfte $\frac{\sqrt{\pi}}{2}$

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$